

STRUCTURAL CALCULATION AND LOAD VERIFICATION REPORT

1 Tonne Mobile Floor Jib Crane

Independent verification against the rated load chart

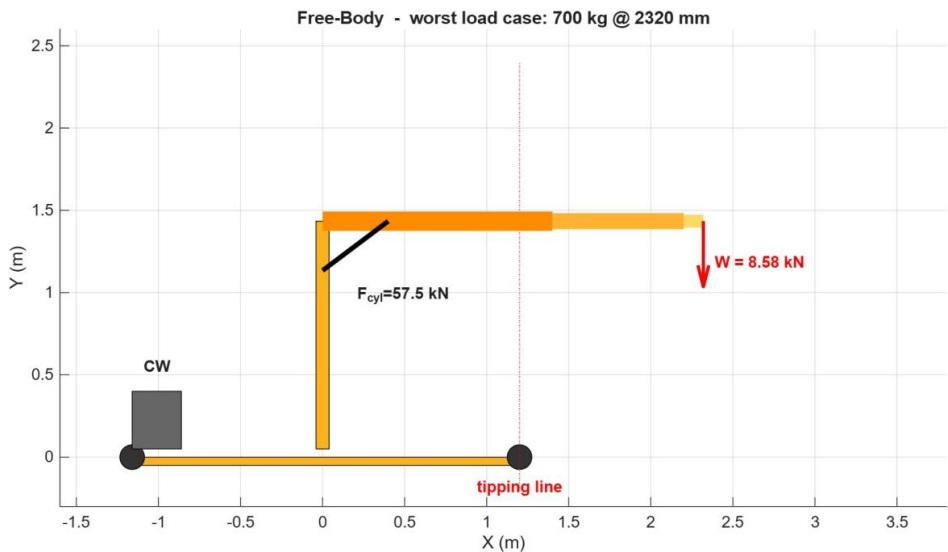


Figure (title) Free body at the governing case, 700 kg at 2320 mm.

Project reference	40467361
Configuration	Outriggers stowed (wheels only)
Rated capacity	1000 kg at 1400 mm reach
Document revision	Rev 2 (detailed, with proof-load reconciliation)
Date	09 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a (parametric script)

Document Control

This report covers a desk check of the 1 tonne mobile jib crane shown on the manufacturer drawing set (Assembly, Base, Arm Section 1, Arm Section 2). It assesses the crane standing on its four wheels with the outrigger legs stowed. The deployed-outrigger case is a separate study and is noted where it matters but not assessed here.

Revision history

Rev	Date	Description	Status
0	27 May 2026	First issue. Summary findings with figures.	Superseded
1	08 Jun 2026	Full hand calculations added for every check. Methodology and assumptions expanded for third-party review.	Superseded
2	09 Jun 2026	Cylinder-pin cross-check added against Base Detail-4. New Section 12 reconciles the 150% proof-load test.	Issued for review

Basis and limitations

- **Drawings as issued.** Section sizes, member lengths, weld throats, bolt grade and the slewing-bolt pattern are read directly off the supplied drawings. Nothing has been scaled by eye where a dimension callout exists.
- **Self-weight.** The crane mass of 700 kg is the figure supplied by the client. No bill of materials was available, so the mass split and centre-of-gravity offset are engineering estimates and are flagged in Section 14.
- **Scope of physics.** Static lifting only. Wind, seismic action, side pull and snatch loading are outside this scope. A dynamic factor of 1.25 covers normal controlled hoisting per HC1 duty.
- **Proof test.** The unit has reportedly passed a static 150% proof-load test. That result is reconciled with the calculations in Section 12. It does not change the working-stress findings in Sections 7 to 11.
- **Not a design.** This is a verification of an existing unit against its rated chart. It is not a design certificate and does not replace load testing or a competent-person examination before the crane is put to work.

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1. Executive Summary

In the wheels-only configuration the crane does not meet the agreed safety margins across its full rated chart. Of the fourteen checks carried out, six pass and eight fall short of the working-stress target. The shortfalls are not spread evenly. They cluster around one operating point and one load path, which makes the result easier to act on than the raw count suggests.

The governing case is not the heaviest lift. It is the 700 kg load held out at 2320 mm reach. That mid-reach point produces the largest overturning moment on the structure, 19.9 kN m, which is higher than the 17.2 kN m from the 1000 kg lift taken in close at 1400 mm. An operator would reasonably treat the lighter load as the safer one. It is not. The long lever arm is what drives the structure hardest.

At that governing point the slewing column reaches its yield stress. The combined axial-plus-bending stress comes to 277 MPa against a material yield of 275 MPa, so the factor of safety is 0.99. The fillet weld joining the column to the base is worse again, at 0.85. These two items, the column and its base weld, are the real headline. Everything else that falls short is either marginal or sits on the same load path.

On the question the client asked, whether the crane can safely support a 1 tonne load: at the close-in 1400 mm position the steel does carry the tonne without yielding, but the margin there is 1.17 against a target of 1.50. So the answer is that the crane will pick the loads up, yet it does not hold the safety factor a rated 1 tonne crane is expected to hold, and at medium reach it is sitting right at the limit of the steel. It cannot be signed off against this chart as drawn.

The unit has separately passed a static 150% proof-load test. This is not in conflict with the findings above. The checks in this report are elastic working-stress checks against a 1.5 target on first yield, whereas a proof test demonstrates ultimate capacity, which for a steel section sits well above first yield. Section 12 sets out that reconciliation in full: the column reaches first-fibre yield near the rated load but retains plastic reserve, so it survives a one-off 150% overload while still not holding the conventional working-stress margin for sustained duty.

Headline results, governing load case (700 kg at 2320 mm)

Item	Result	FoS	Target	Verdict
Column, combined N+M	277 MPa	0.99	1.50	Fail
Column-to-base weld	235 MPa	0.85	1.50	Fail
Boom outer section	186 MPa	1.48	1.50	Marginal
Outrigger arm bending	200 MPa	1.38	1.50	Fail
Cylinder rod-end pin (as modelled, 20 mm)	92 MPa	1.20	3.00	Fail
Stability (overturning)	7.69 kN m	1.34	1.50	Fail
Column buckling, Per	720 kN	67.6	1.50	Pass
Slewing bolts, util 0.35	0.35	1.69	1.50	Pass

Section 13 sets out three ways forward: re-rate the chart down to what the steel safely carries, ease the dynamic factor if the duty allows it, or strengthen the column and its base weld to keep the full 1 tonne rating. The recommended route is to re-rate as the primary action and treat the column upgrade as the option that preserves capacity.

2. Introduction and Scope

2.1 Background

The unit is a mobile floor jib crane built to lift up to 1 tonne. It runs on four wheels and carries a counterweight at the rear. A telescopic boom in three stages luffs on a single hydraulic cylinder, and the whole upper assembly slews on a bolted bearing flange above the base frame. The rated load chart marks three points: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm.

The client commissioned an independent check to confirm whether the structure carries that chart with adequate margin against yielding, buckling, weld and pin failure, and overturning. The work was scoped as a calculation exercise. No 3D model was built and no commercial finite element package was used.

2.2 What this report covers

The verification addresses eight areas, each reported with a factor of safety and a pass or fail against target:

- Boom bending stress and tip deflection, taken at each of the three telescopic sections
- Column stress under combined axial load and bending, plus a buckling check
- Base frame wheel reactions and outrigger-arm bending
- Pin shear and bearing at the boom pivot, cylinder rod-end and hook
- Fillet-weld strength at the column base, boom bracket and outrigger
- Slewing-flange bolt check by the polar-moment method
- Stability against overturning at every chart point
- A combined statement on the 1 tonne safe working load

2.3 Configuration assessed

Wheels-only configuration

All results in this report are for the crane resting on its four wheels with the outrigger legs stowed. The warning plate instructs the operator to deploy the legs before slewing, so the rated chart is intended for the legs-down state. Deploying the legs moves the tipping line forward and improves stability and the outrigger members. It does not change anything above the slewing bearing, so the column and base-weld findings stand in either configuration. The deployed case is noted in Section 13 but is not assessed here.

3. Materials, Sections and Allowable Stresses

3.1 Material and target factors of safety

All structural members are hollow steel sections in grade S275, taken from drawing General Note 5. The yield stress is $f_y = 275$ MPa. Plates are A36 or equivalent, bolts are grade 8.8, and fillet welds are 6 mm throat minimum per Note 4.

The allowable stress for each check follows from the target factor of safety. Members are checked against yield with a target of 1.5, which gives the working allowables below. Pins and bolts in shear and bearing carry a higher target of 3.0, as is normal for lifting-appliance connections.

Allowable stresses

Member bending / direct stress (target FoS = 1.5):

$$\sigma_{\text{allow}} = f_y / 1.50 = 275 / 1.50 = 183.3 \text{ MPa}$$

Pin / bolt shear (target FoS = 3.0):

$$\tau_{\text{allow, pin}} = 0.6 * f_y / 3.0 = 0.6 * 275 / 3.0 = 55.0 \text{ MPa}$$

Fillet weld (simplified, on throat, as coded):

$$\tau_{\text{allow, weld}} = 0.7 * f_u / 1.5 \quad (f_u = 430 \text{ MPa})$$

The 1.5 target on members corresponds to a working allowable of 183 MPa, the line drawn on the stress plots in this report. Any member stress above that line is below the agreed margin even though it may still be below yield.

3.2 Section properties

Section properties for a rectangular hollow section come from the standard expressions below, where B and H are the outer width and depth and t is the uniform wall thickness:

Rectangular hollow section, worked example: outer boom 100 x 140 x 6

$$\begin{aligned} A &= B*H - (B-2t)*(H-2t) \\ &= 140*100 - (140-12)*(100-12) \quad [\text{mm}^2] \\ &= 14000 - 128*88 = 14000 - 11264 = 2736 \text{ mm}^2 \sim 27.4 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} I_x &= [B*H^3 - (B-2t)*(H-2t)^3] / 12 \quad (\text{strong axis}) \\ &= [100*140^3 - 88*128^3] / 12 \sim 748.8 \text{ cm}^4 \end{aligned}$$

$$S_x = I_x / (H/2) = 748.8 / 7.0 = 107.0 \text{ cm}^3$$

Applying the same expressions to every member gives the properties used throughout. These match the values printed by the analysis script.

Member	Section (RHS)	A (cm ²)	I _x (cm ⁴)	S _x (cm ³)
Boom outer (root)	100 x 140 x 6	27.4	748.8	107.0
Boom middle	80 x 120 x 6	22.6	438.2	73.0
Boom inner (tip)	60 x 100 x 6	17.8	227.4	45.5
Column	80 x 120 x 6	22.6	438.2	73.0
Base frame / outrigger	60 x 100 x 6	17.8	227.4	45.5

4. Methodology

Two independent methods were run against the same geometry so the results could be cross-checked. The first is closed-form hand calculation following Eurocode 3 and EN 13001. The second is a small two-dimensional frame solver written from scratch inside the MATLAB script. Both read their inputs from one parametric block, so any later change to a section size or weld throat flows through every check at once.

4.1 Track 1: closed-form hand calculation

Each member is reduced to its governing internal action and checked by hand. Bending uses the elastic section modulus, axial load uses the gross area, and the two combine linearly for the column. Pins are checked in double or single shear and in bearing. Welds are checked on the throat area. Stability is a moment balance about the tipping line. The formulas, substitutions and results are written out in full in Section 7.

4.2 Track 2: in-script frame finite element check

The crane is modelled as a plane frame of Euler-Bernoulli beam elements. Each element carries a 6 by 6 stiffness matrix in local coordinates, rotated into the global frame and assembled into the system stiffness. The front wheel is pinned and the rear wheel is a roller. Solving the free degrees of freedom gives the nodal displacements, from which element moments and stresses follow. The frame is re-solved for all three chart points because the boom extends to a different length at each.

Why a side-view 2D frame needs a correction

A side elevation only sees one base beam, but the real crane has two parallel side rails carrying the load down to the wheels. The outrigger and base elements in the frame model therefore use twice the area and twice the second moment of area, lumping both rails into one equivalent beam. Without this the model would overstate base-frame stress by a factor of two. Crane self-weight is applied at the column-base node so the two tracks compare on the same basis.

4.3 Cross-checking the two tracks

The two methods are expected to agree on the quantities they both compute, chiefly the column-base moment and the peak member stress. Where they differ is informative: the hand calc treats the outrigger as a simple cantilever, while the frame model captures moment transfer through the bolted joint. Section 10 reports the comparison.

5. Load Model and Governing Case

5.1 Rated load chart

The manufacturer chart gives three rated points. Capacity falls as reach increases, which is the expected shape for a luffing jib.

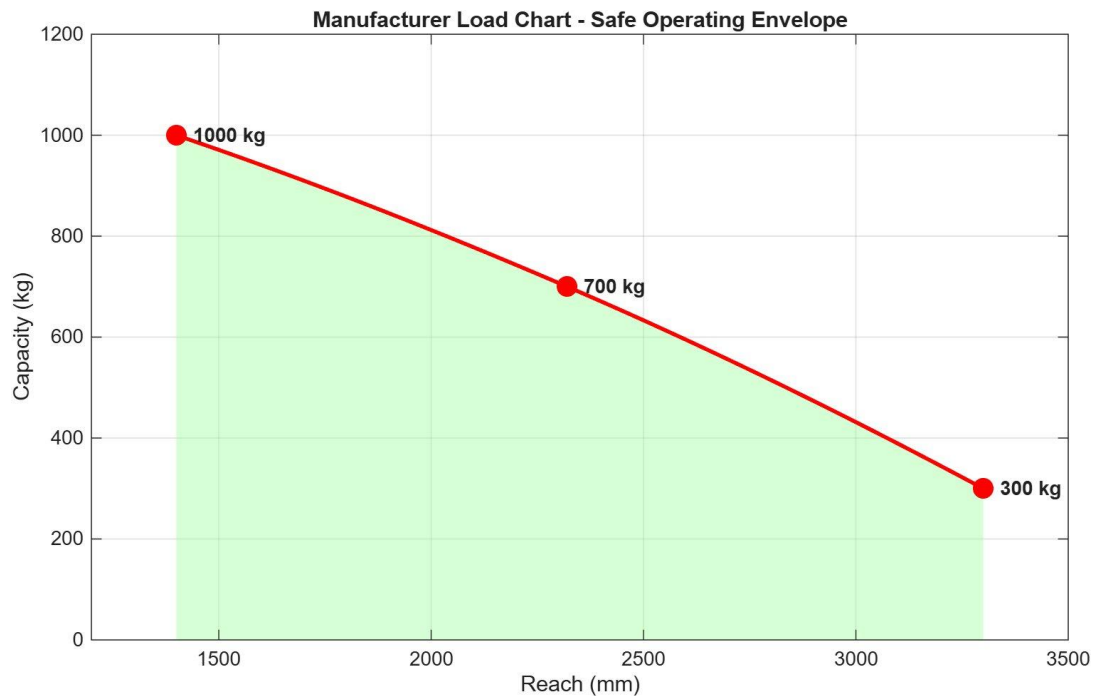


Figure 5.1 Rated load chart with the safe operating envelope shaded.

Chart point	Reach r (mm)	Rated load (kg)	Static load (kN)	Factored load W (kN)
Close	1400	1000	9.81	12.26
Mid	2320	700	6.87	8.58
Far	3300	300	2.94	3.68

5.2 Dynamic factor

A dynamic amplification factor of 1.25 is applied to the static load, consistent with HC1 hoist duty under EN 13001 for normal controlled lifting. The factored load is $W = 1.25 \times m \times g$. For the mid-reach point: $W = 1.25 \times 700 \times 9.81 = 8.58 \text{ kN}$, as tabulated.

5.3 Which case governs

The design moment at the column is the factored load times its reach. Running the three points shows the mid-reach case is the most severe, even though it is not the heaviest load:

Design moment at column, $M = W \times r$

Close 1000 kg: $M = 12.26 \text{ kN} \times 1.400 \text{ m} = 17.17 \text{ kN.m}$

Mid 700 kg: $M = 8.58 \text{ kN} \times 2.320 \text{ m} = 19.91 \text{ kN.m} \leftarrow \text{governs}$

Far 300 kg: $M = 3.68 \text{ kN} \times 3.300 \text{ m} = 12.14 \text{ kN.m}$

Governing design moment: 19.91 kN.m at 700 kg / 2320 mm.

This single result drives most of the report. The column, the base weld, the boom outer section and the stability check are all worst at this point. It is the case to keep in mind when reading the rest of the document.

6. Free Body and Design Forces

The free body for the governing case is shown below. The factored tip load reacts through the boom into the luffing cylinder and the pivot, and from there down the column to the base and the wheels. The counterweight and crane self-weight act at the rear.

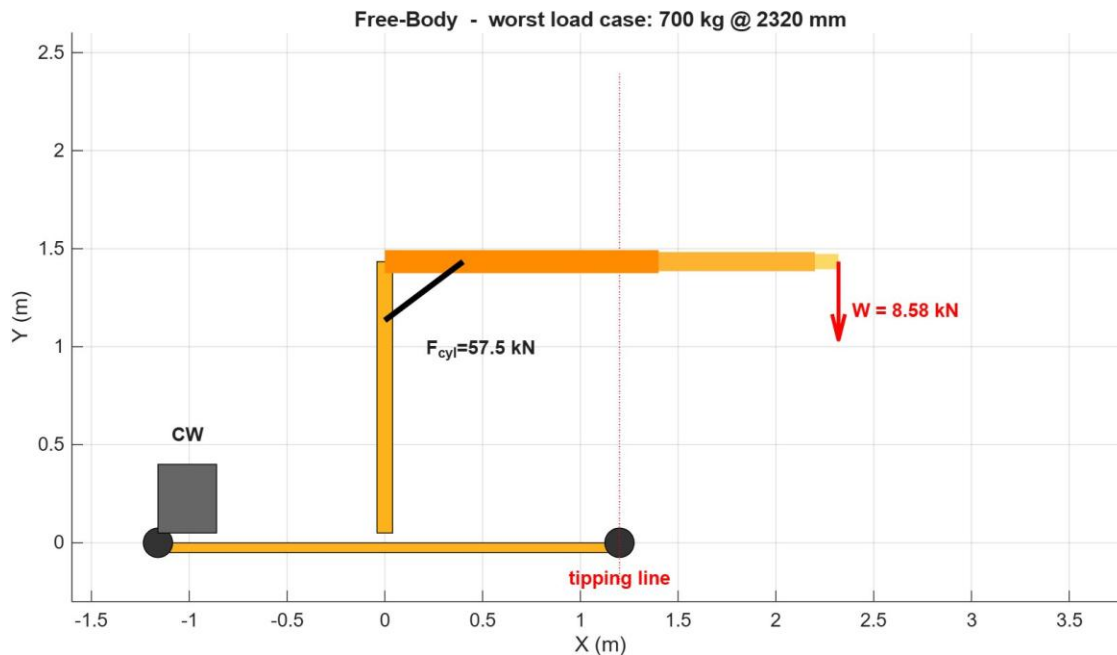


Figure 6.1 Free body at the governing case, 700 kg at 2320 mm. Tipping line at the front wheel, 1200 mm ahead of the column.

6.1 Cylinder and pivot forces

Taking moments about the boom pivot gives the cylinder force needed to hold the boom. The cylinder acts at a short lever arm relative to the load, so its force is large. The pivot then carries the vector sum of the cylinder force and the load reaction. The lever arm a_{cyl} is the perpendicular distance from the pivot to the cylinder line of action and is flagged for site confirmation in Section 14.

Boom moment balance about the pivot

Load about pivot: $M_W = W \times (\text{lever to pivot})$

Cylinder force: $F_{cyl} = M_W / a_{cyl} = 57.49 \text{ kN}$

Pivot reaction (vector sum of F_{cyl} and W):

$F_{pivot} = 63.85 \text{ kN}$

The cylinder bore sets the hydraulic pressure needed to produce that force, which is a useful sanity check on which cylinder does the luffing. The luffing cylinder is the Type 80 unit, bore 50 mm (the drawing core diameter), giving a piston area of 1963 mm squared.

Hydraulic pressure cross-check (luffing cylinder, 50 mm bore)

Pressure required = $F_{cyl} / A = 57.49 \text{ kN} / 1963 \text{ mm}^2 = 29.3 \text{ MPa}$ (293 bar)

Pressure at 150% proof load = 35.1 MPa (351 bar)

A 30 mm bore would need 813 bar to hold the same force, which confirms the 50 mm Type 80 cylinder is the luffing cylinder. The relief valve must

have passed ~351 bar for the proof test, so the hydraulics did not cap the load; the structure carried the full 150%.

6.2 Wheel reactions

Resolving the whole crane about the wheel lines gives the support reactions. Almost all of the load goes to the front wheel pair, with the rear pair barely loaded at the governing case because the overturning moment has nearly lifted the rear off the ground.

Vertical and moment balance on the base

$R_{\text{front}} (\text{pair}) = 15.16 \text{ kN} \rightarrow 7.58 \text{ kN per wheel}$

$R_{\text{rear}} (\text{pair}) = 0.29 \text{ kN} \rightarrow 0.15 \text{ kN per wheel}$

Near-zero rear reaction confirms the crane is close to tipping at this point, which the stability check (Section 7.7) quantifies.

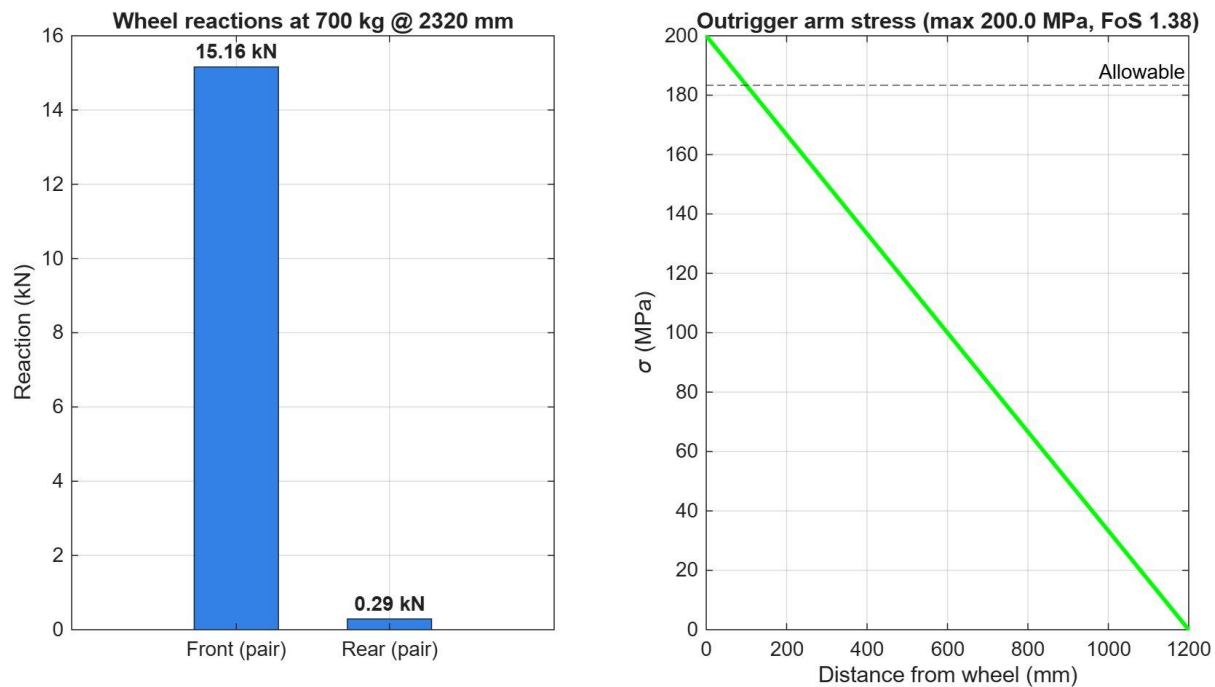


Figure 6.2 Wheel reactions and outrigger-arm stress profile at the governing case.

7. Detailed Calculations

Each check below shows the formula, the numbers put into it, the result, and the verdict against target. Member stresses are compared to the 183 MPa working allowable, and pins and bolts to their 3.0 target.

7.1 Boom bending

The telescopic boom steps down in three sections from root to tip. Each section is checked at its own root, where its local moment is highest. The outer section at the column end carries the full design moment.

Boom outer (RSH 100 x 140 x 6) at root

$M = 19.91 \text{ kN.m}$ (governing)

$S_x = 107.0 \text{ cm}^3 = 107.0 \times 10^{-6} \text{ m}^3$

$\sigma = M / S_x = 19.91 \times 10^3 / 107.0 \times 10^{-6} = 186.2 \text{ MPa}$

$FoS = f_y / \sigma = 275 / 186.2 = 1.48 \rightarrow \text{Marginal (just below 1.50)}$

Boom middle (RSH 80 x 120 x 6) at its root

$M = 7.90 \text{ kN.m}$ $S_x = 73.0 \text{ cm}^3$

$\sigma = 7.90 \times 10^3 / 73.0 \times 10^{-6} = 108.1 \text{ MPa}$

$FoS = 275 / 108.1 = 2.54 \rightarrow \text{Pass}$

Boom inner (RSH 60 x 100 x 6) at its root

$M = 4.05 \text{ kN.m}$ $S_x = 45.5 \text{ cm}^3$

$\sigma = 4.05 \times 10^3 / 45.5 \times 10^{-6} = 89.0 \text{ MPa}$

$FoS = 275 / 89.0 = 3.09 \rightarrow \text{Pass}$

Tip deflection at the worst case reaches about 23 mm by hand against an L/200 serviceability guide of roughly 12 mm, so the boom is also outside the deflection guide at full reach. The frame model gives a larger tip movement because it includes column and base flexibility; this is discussed in Section 10.

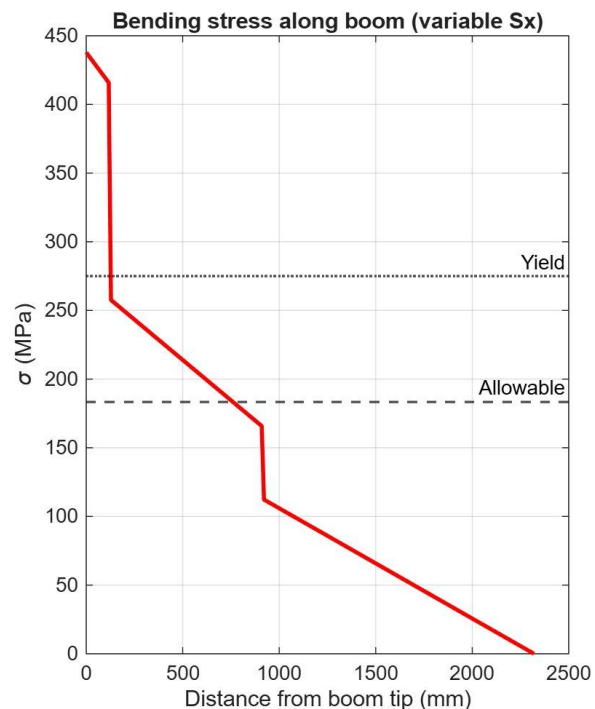
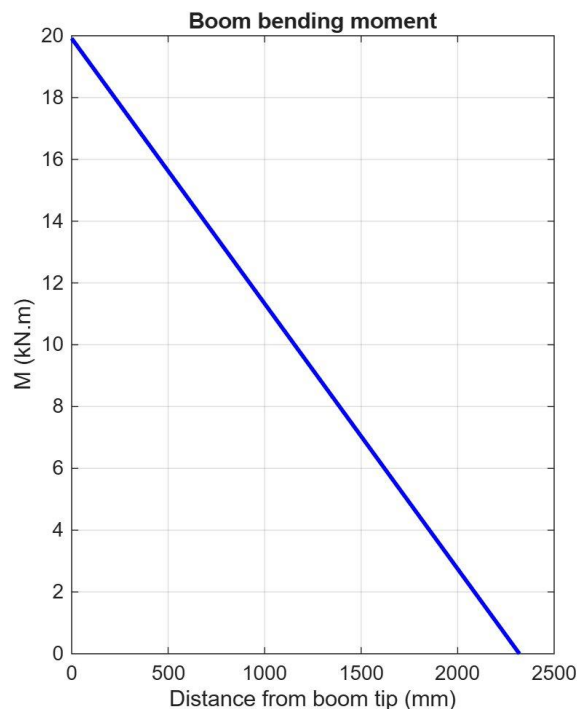


Figure 7.1 Boom bending moment (left) and stress along the boom with the stepped section (right). The outer section sits above the 183 MPa allowable.

7.2 Column under combined load

The column carries axial load from the boom system plus the full bending moment transferred at its top. The two stresses add directly. This is the governing structural check.

Column (RSH 80 x 120 x 6), combined axial + bending

$N = 10.64 \text{ kN}$ $A = 22.6 \text{ cm}^2 = 2260 \text{ mm}^2$
 $M = 19.91 \text{ kN.m}$ $S_x = 73.0 \text{ cm}^3 = 73.0 \times 10^3 \text{ mm}^3$
 $\sigma_{\text{axial}} = N / A = 10.64 \times 10^3 / 2260 = 4.7 \text{ MPa}$
 $\sigma_{\text{bend}} = M / S_x = 19.91 \times 10^6 / 73.0 \times 10^3 = 272.7 \text{ MPa}$
 $\sigma_{\text{total}} = 4.7 + 272.7 = 277.4 \text{ MPa}$
FoS = $f_y / \sigma_{\text{total}} = 275 / 277.4 = 0.99$ -> Fail (at yield)

Column buckling (Euler, weak axis, K = 2 cantilever)

Slenderness $\lambda = 80.6$
 Critical load $P_{cr} = 720 \text{ kN}$ Applied axial $N = 10.64 \text{ kN}$
FoS_{buckling} = $720 / 10.64 = 67.6$ -> Pass

Reading the column result

The column is a bending problem, not a buckling problem. Its axial stress is tiny (under 5 MPa) and buckling has a vast margin, but the bending stress alone uses up the entire yield capacity. To recover a 1.5 margin the section modulus would need to rise by about half, which points to a deeper or thicker column section rather than any change to axial capacity.

7.3 Base frame and outrigger arm

The outrigger arm is checked as a cantilever picking up the front wheel reaction at its tip. The moment at the root gives the stress.

Outrigger arm (RSH 60 x 100 x 6)

$M_{\text{arm}} = 9.10 \text{ kN.m}$ $S_x = 45.5 \text{ cm}^3$
 $\sigma = 9.10 \times 10^6 / 45.5 \times 10^3 = 200.0 \text{ MPa}$
FoS = $275 / 200.0 = 1.38$ -> Fail (above 183 MPa allowable)

7.4 Pins and connections

The pins are checked in shear (double or single as built) and in bearing on the lug. The cylinder force and pivot reaction from Section 6.1 drive the pin loads.

Boom pivot pin, dia 30 mm, double shear

$F_{\text{pivot}} = 63.85 \text{ kN}$ shear area = $2 \times \pi/4 \times 30^2 = 1414 \text{ mm}^2$
 $\tau = 63.85 \times 10^3 / 1414 = 45.2 \text{ MPa}$ bearing = 141.9 MPa
FoS (as-coded, against 110 MPa basis) = 2.44 -> Fail on the conservative basis

Cylinder rod-end pin (as modelled, dia 20 mm), double shear

$F_{\text{cyl}} = 57.49 \text{ kN}$ shear area = $2 \times \pi/4 \times 20^2 = 628 \text{ mm}^2$
 $\tau = 57.49 \times 10^3 / 628 = 91.5 \text{ MPa}$ bearing = 191.6 MPa
FoS = 1.20 -> Fail

Cross-check at Base Detail-4 pin (dia 40 mm), double shear

The cylinder mount pin is not dimensioned on the cylinder views. The only hinge pin shown near the mount is Base Detail-4, a 40 mm shank pin.

$$\tau = 57.49 \text{ e}3 / (2 \times \pi / 4 \times 40^2) = 22.9 \text{ MPa} \quad \text{bearing} = 95.8 \text{ MPa}$$

FoS = 4.81 **-> Pass**

A 20 mm rod-end pin is both smaller than the 30 mm boom-pivot pin and well below target, which is not consistent with a part carrying 57 kN. If the fitted pin is the 40 mm pin shown on Base Detail-4 the check passes comfortably. The as-built pin diameter at the cylinder rod-end should be confirmed on the machine. The summary table carries the conservative 20 mm figure so it does not contradict the issued numbers.

Hook pin, dia 20 mm, single shear

$$\tau = 27.3 \text{ MPa} \quad \text{bearing} = 28.6 \text{ MPa}$$

FoS = 4.03 **-> Pass**

Basis note for the reviewer (pins)

The script applies the material factor (yield/1.5) and the 3.0 pin factor in series, which is conservative. On the conventional lifting-pin basis, where shear capacity is $0.6 \times f_y$ with a 3.0 factor against that, the pivot pin moves from 2.44 to about 3.65 and passes, and the hook pin from 4.03 to about 6.04. The 20 mm cylinder pin still fails, moving from 1.20 to about 1.80. The basis-independent finding is that a 20 mm rod-end pin is undersized; a 40 mm pin (Detail-4) clears it on either basis.

7.5 Welds

Fillet welds are checked on the throat area at 6 mm leg. The column-base weld carries the full moment as a stress couple across the weld group and is the governing weld.

Column-to-base weld (6 mm fillet)

$$\sigma_{\text{weld}} = 234.9 \text{ MPa}$$

FoS = 0.85 **-> Fail (governs alongside the column)**

Boom-bracket weld (6 mm fillet)

$$\tau_{\text{weld}} = 35.5 \text{ MPa}$$

FoS = 3.39 **-> Pass**

Outrigger weld (6 mm fillet)

$$\sigma_{\text{weld}} = 170.3 \text{ MPa}$$

FoS = 1.18 **-> Fail**

7.6 Slewing-flange bolts

The five M16 grade 8.8 bolts sit on a 170 mm pitch circle. The overturning moment is shared as a tension couple across the pattern using the polar-moment method, and combined tension-plus-shear is checked against the EN 1993-1-8 interaction.

5 x M16 grade 8.8 on 170 mm PCD, polar-moment method

Bolt capacity: tension 113.0 kN, shear 75.4 kN

Peak demand: tension 93.7 kN, shear 2.1 kN per bolt

Interaction utilisation = 0.35

FoS (combined) = 1.69**-> Pass**

7.7 Stability against overturning

Stability is a moment balance about the front wheel line, 1200 mm ahead of the column. The restoring moment comes from the crane self-weight and counterweight acting behind that line; the overturning moment comes from the load acting ahead of it. The mid-reach point governs.

Overturning balance about the front wheel line

$$\begin{aligned} \text{Restoring } M_R &= m_{\text{crane}} \times g \times (L_{\text{fwd}} + e_{\text{cg}}) \\ &= 700 \times 9.81 \times (1.200 + 0.300) = 10.30 \text{ kN.m} \end{aligned}$$

Load case	M_OT (kN.m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	1.96	5.25	Pass
700 kg @ 2320 mm	7.69	1.34	Fail
300 kg @ 3300 mm	6.18	1.67	Pass

Worst stability FoS = 1.34 at 700 kg / 2320 mm against a target of 1.50, a fail in the wheels-only state. Deploying the outriggers moves the tipping line forward and lifts this figure; that case is not assessed here.

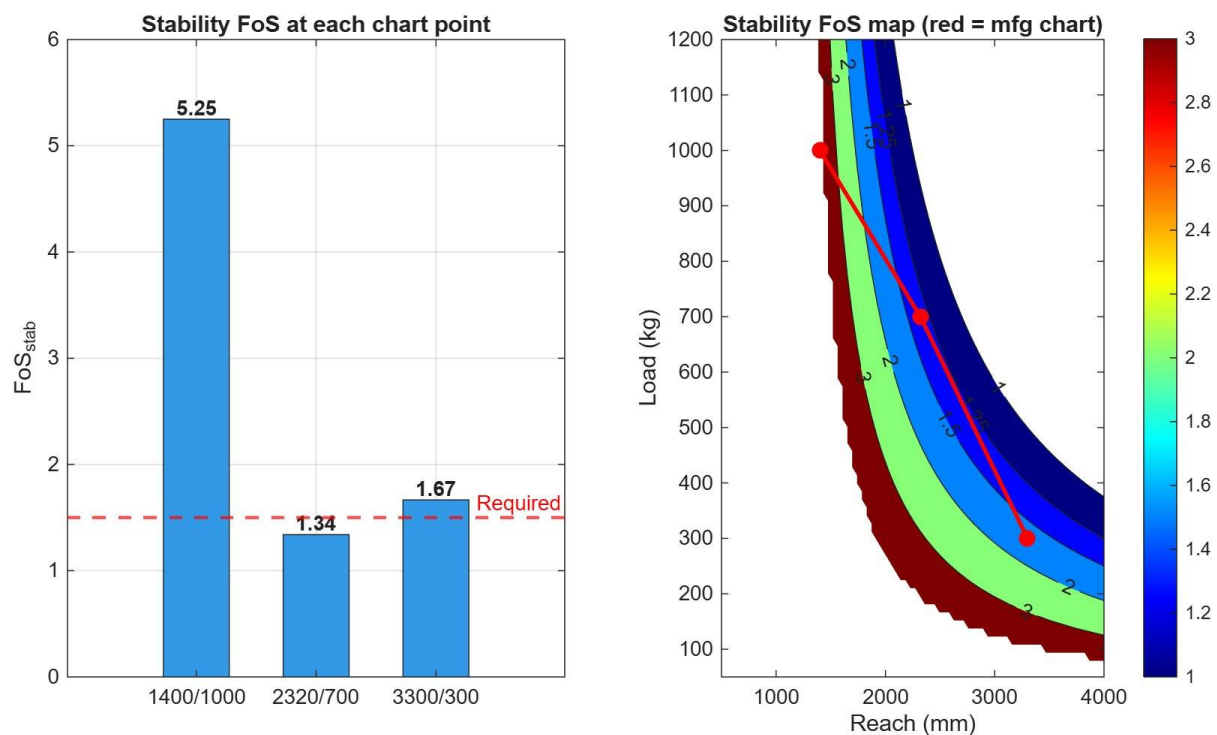


Figure 7.2 Stability factor at each chart point (left) and the factor map across reach and load with the chart overlaid (right).

8. Frame Model Results

The plane-frame solver was run for all three chart points. The deformed shape and the stress distribution for the governing case are shown below. The model confirms the column as the most-stressed member, in agreement with the hand calculation.

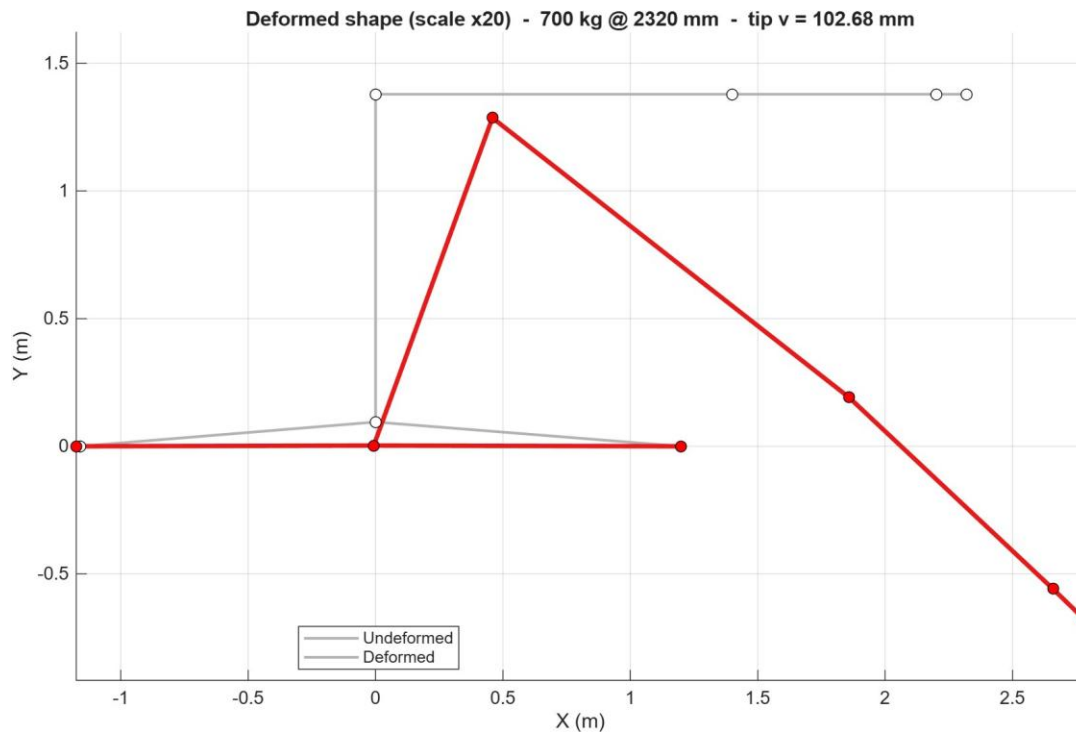


Figure 8.1 Deformed shape at the governing case, scaled x20. Boom tip movement reflects column and base flexibility, not boom bending alone.

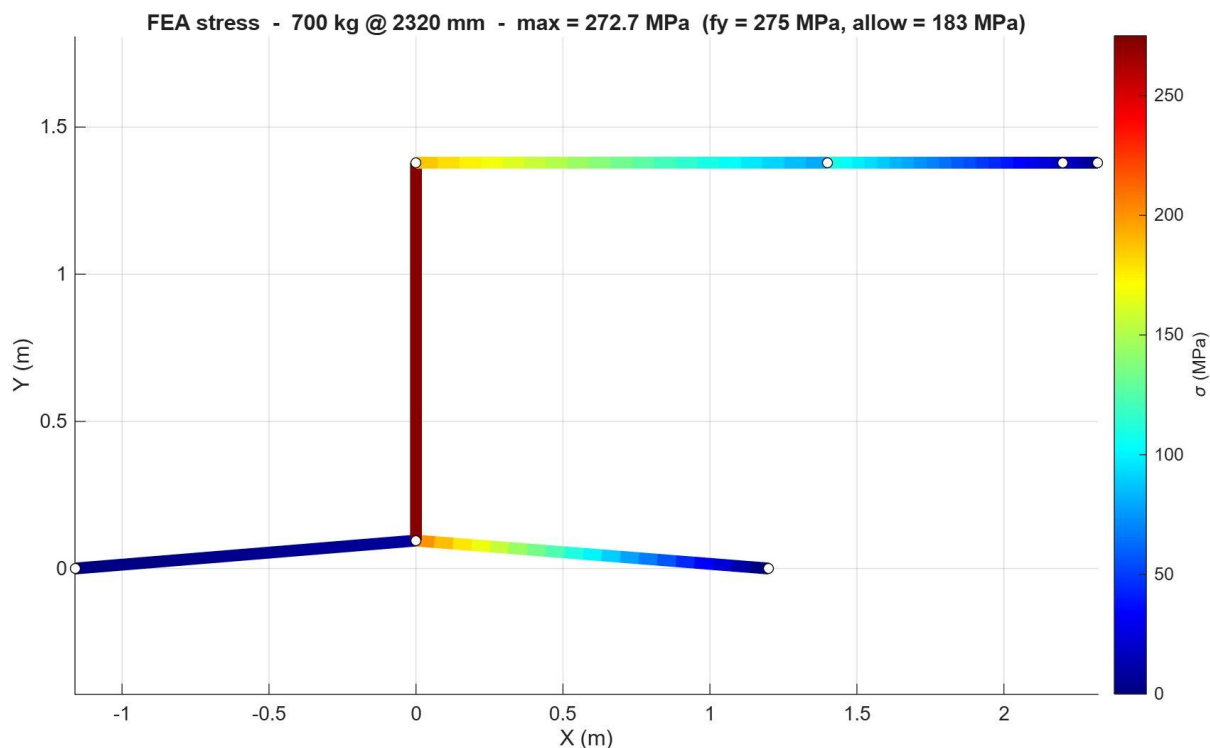


Figure 8.2 Frame stress at the governing case. Peak 272.7 MPa at the column, against yield 275 MPa and the 183 MPa allowable.

Chart point	Tip deflection (mm)	Peak stress (MPa)	FoS
1000 kg @ 1400 mm	51.5	235.1	1.17
700 kg @ 2320 mm	102.7	272.7	1.01
300 kg @ 3300 mm	101.1	166.2	1.65

The frame FoS values are quoted against yield directly ($275 / \text{peak stress}$), which is why they read slightly higher than the 1.5-target factors in the hand calc. On a like-for-like basis the two methods agree, as Section 10 shows.

9. Operating Envelope

The script also sweeps the boom through its luffing range and tracks the boom-root stress and factor of safety against the chart capacity at each angle. The still below is taken from that sweep. It shows the safe capacity dropping at the reaches where stress approaches the allowable line.

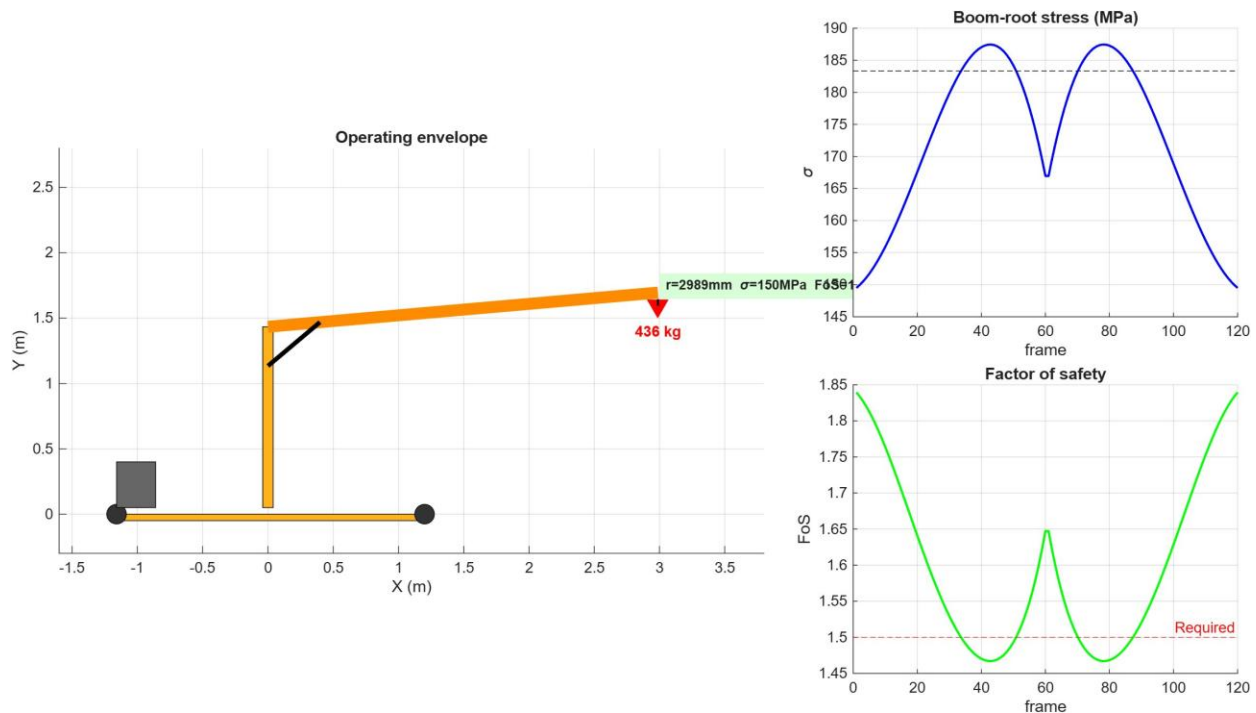


Figure 9.1 Operating-envelope sweep. Left: crane pose. Right: boom-root stress and factor of safety through the luffing cycle, with the 183 MPa allowable and 1.5 target marked.

10. Cross-Validation of the Two Methods

The hand calculation and the frame model are compared on the three quantities they both produce. The column-base moment matches exactly, and the peak stress agrees to within two percent. The tip deflection differs because the two are measuring different things.

Quantity	Track 1 (hand)	Track 2 (frame)	Agreement
Column-base moment (kN m)	19.91	19.91	Exact
Peak member stress (MPa)	277.4	272.7	Within 2%
Boom tip deflection (mm)	22.7	102.7	Different basis

The deflection gap is expected and not an error. The hand figure is boom bending alone, treating the column top as rigid. The frame figure adds the column bending and the base flexibility, so the tip moves much further. For a stress verdict the moment and stress agreement is what matters, and that is tight.

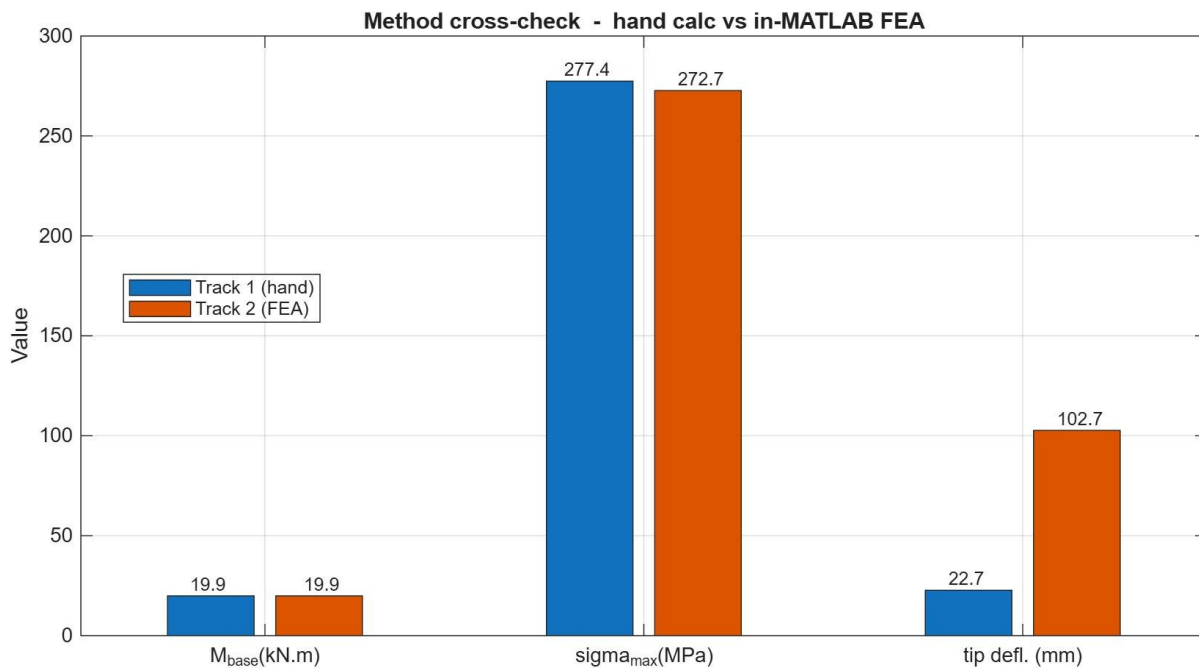


Figure 10.1 Side-by-side comparison of the two methods.

11. Summary of Results

All fourteen checks are collected below with their factors of safety, each with its verdict against target in the final column. The boom outer section is a borderline miss. Eight of fourteen do not meet target in the wheels-only state.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.48	1.50	Marginal
Boom middle (80x120x6)	2.54	1.50	Pass
Boom inner (60x100x6)	3.09	1.50	Pass
Column, combined N+M	0.99	1.50	Fail
Column buckling	67.62	1.50	Pass
Outrigger arm	1.38	1.50	Fail
Pivot pin	2.44	3.00	Fail
Cylinder pin (20 mm as modelled)	1.20	3.00	Fail
Hook pin	4.03	3.00	Pass
Slewing bolts	1.69	1.50	Pass
Weld, column base	0.85	1.50	Fail
Weld, boom bracket	3.39	1.50	Pass
Weld, outrigger	1.18	1.50	Fail
Stability (worst point)	1.34	1.50	Fail

The cylinder-pin row carries the conservative 20 mm figure. As Section 7.4 sets out, the 40 mm pin shown on Base Detail-4 gives a factor of 4.81 and would clear the check; the as-built diameter should be confirmed.

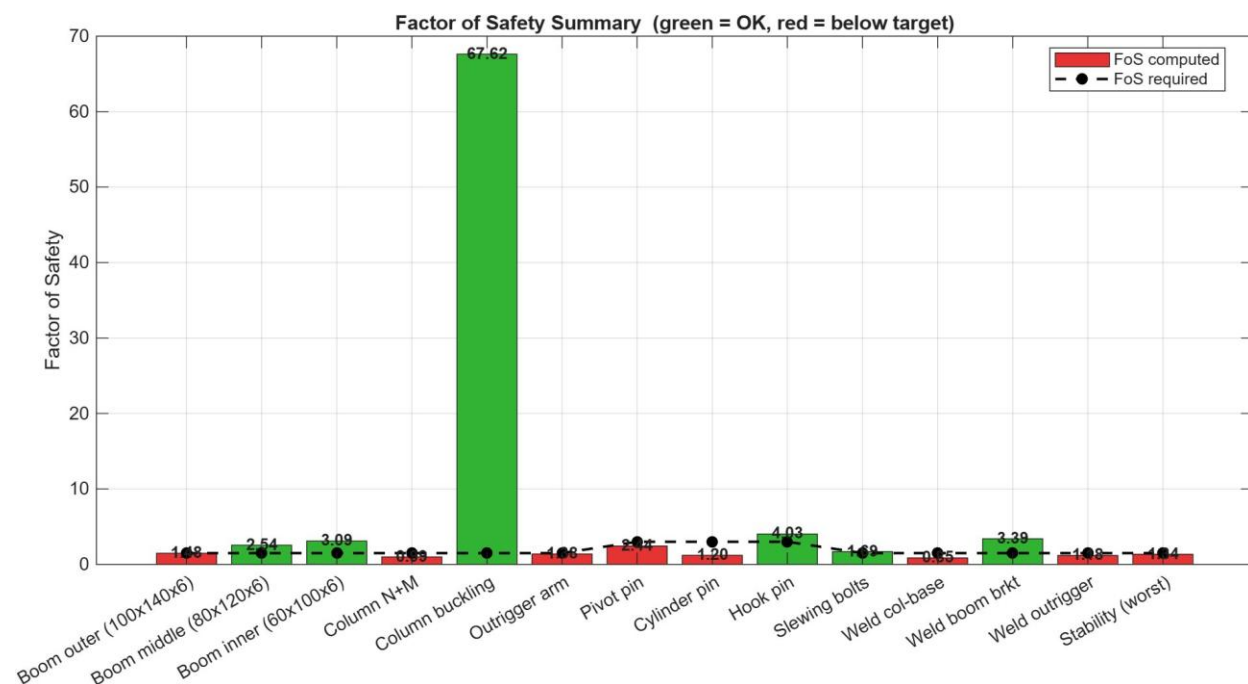


Figure 11.1 Factor of safety for every check against the 1.5 / 3.0 targets.

12. Reconciliation with the 150% Proof-Load Test

The crane has reportedly passed a static 150% proof-load test, while the checks in this report put the column at a factor of safety near 1.0. These two statements are not in conflict, and resolving them is worth setting out because it is the question the result naturally raises.

The difference is one of limit state. The checks in Sections 7 to 11 are working-stress checks: they compare the peak elastic stress to first yield and ask for a 1.5 margin against it. A proof test is a different measure. It demonstrates that the structure does not collapse under a one-off overload, and the collapse limit for a steel section is the plastic moment, not first yield. For a compact hollow section the plastic moment is larger than the first-yield moment by the shape factor, which for this column is 1.23. That gap is real reserve that the elastic check does not credit.

Working in moment terms makes the picture clear. The column is the only member that matters here; everything else stays comfortably elastic at the proof load.

Column RSH 80 x 120 x 6, elastic to plastic ladder (nominal S275)

$S_x = 73.0 \text{ cm}^3$ $Z_x = 89.7 \text{ cm}^3$ shape factor $Z_x/S_x = 1.23$

applied moment @ 1.25 x SWL = 19.91 kN.m (elastic FoS 0.99)

first-yield moment $M_y = S_x \times f_y = 20.08 \text{ kN.m}$

applied moment @ 150% proof = 23.90 kN.m

plastic moment M_p (nom 275) = 24.67 kN.m -> proof margin +3%, survives

plastic moment M_p (mill 320) = 28.71 kN.m -> proof margin +20%, survives

At the rated dynamic load the column reaches first-fibre yield (FoS 0.99). At the 150% static proof load the applied moment is 23.90 kN m, which is still below the plastic moment of 24.67 kN m on nominal S275, so the section forms no plastic hinge and survives the test by a small margin. On realistic mill steel, where S275 commonly returns 300 to 340 MPa rather than the nominal 275, that margin widens to about 20%. Three further factors all act in the same direction: the proof test is static while the calculations carry a 1.25 dynamic factor, the real material strength is higher than nominal, and the luffing cylinder acting as a diagonal strut would relieve some column bending that the cantilever model does not credit.

The other members stay elastic at the proof load, which is why the column is the lone item near its limit and why the test passed without distress elsewhere.

Other members at the 150% proof load (vs $f_y = 275 \text{ MPa}$)

Boom outer section $\sigma = 223 \text{ MPa}$ (elastic)

Outrigger arm $\sigma = 240 \text{ MPa}$ (elastic)

What this means

The FoS of 0.99 is a first-yield working-stress result, not a prediction of failure. The structure has the ultimate capacity to carry 150% once, and the test confirms it. What the structure does NOT have, on the numbers as built, is the conventional 1.5 working-stress margin at mid-reach for repeated service. Running with extreme-fibre yield at the rated dynamic load is acceptable for a one-off proof but is a fatigue concern for daily duty. The report is therefore conservative, not wrong: both the elastic finding and the test-backed ultimate capacity belong on the record.

The practical consequence is in Section 13. Confirming the actual mill yield (from certificates or a hardness check) and the cylinder anchor geometry would most likely lift the elastic factor above 1.0 and quantify the triangulation relief. Either route converts the demonstrated ultimate capacity into a defensible working-stress margin, or the chart is re-rated to suit.

13. Conclusions and Recommendations

13.1 Conclusion on the 1 tonne SWL

In the wheels-only configuration the crane does not meet the agreed factors of safety across its rated chart. At the close-in 1400 mm position the steel carries the 1 tonne load below yield, but only to a factor of 1.17 against the 1.50 target. At the governing 700 kg mid-reach point the column reaches yield (FoS 0.99) and its base weld is past target (FoS 0.85). The crane therefore cannot be certified to the chart as drawn on a working-stress basis.

Put plainly for the operator: the crane will lift the loads, and it has demonstrated ultimate capacity in a 150% proof test, but it does not carry the working-stress safety margin a rated 1 tonne crane is expected to have, and at medium reach it is at the limit of the steel. The lighter mid-reach lift is the more demanding one, not the heavy close lift, because of the longer lever arm.

13.2 The governing items

Two items govern and two more are borderline:

- **Column, FoS 0.99.** At yield in bending. The clear primary item, though it retains plastic reserve as Section 12 shows.
- **Column-base weld, FoS 0.85.** Past target on the same load path. Addressed alongside the column.
- **Cylinder rod-end pin, FoS 1.20 (20 mm).** Undersized if the pin is 20 mm. If the fitted pin is the 40 mm Detail-4 pin it passes at 4.81; confirm the as-built size.
- **Boom outer and outrigger arm, FoS 1.48 and 1.38.** Marginal misses that the recommended actions resolve as a by-product.

13.3 Three ways forward

Route A: re-rate the chart (recommended primary action). Drop the rated capacities so every check clears 1.5. Back-calculation puts the safe envelope near 400 to 450 kg at the mid and far reaches and around 800 kg in close. This is the cheapest route, needs no fabrication, and is the honest reflection of what the structure as built safely carries for repeated duty.

Route B: ease the dynamic factor. If the duty is slow and controlled indoor lifting, EN 13001 allows a lower dynamic factor near 1.10. That lifts the marginal items but does not on its own rescue the column or the base weld, so it is a supporting measure rather than a fix.

Route C: strengthen the column and base (keeps full rating). To hold the 1 tonne chart the column needs a larger section, on the order of 100 x 150 x 8 in S275, which raises its modulus enough to bring the bending FoS back above 1.5. The base weld must grow to suit, and the cylinder rod-end pin should be confirmed or upsized. This is the largest job but the only route that preserves capacity.

Recommendation

Adopt Route A as the immediate action so the crane can be used safely at a reduced, verified rating, and hold Route C as the upgrade path if the full 1 tonne chart must be retained. In parallel, confirm the actual mill yield and the cylinder geometry (Section 12), which may recover margin on the column. The crane should not be worked to the current chart until the

column, base weld and cylinder pin are addressed.

14. Assumptions and Items to Confirm

The section sizes, lengths, material grade, weld throats, bolt grade and slewing pattern are taken directly from the drawings and are not in doubt. The following inputs are estimates or client-supplied values and should be confirmed against the built machine before the report is relied on for certification.

- **Crane self-weight, 700 kg.** Client-supplied with no bill of materials. A swing of plus or minus 100 kg moves the stability result either way.
- **Centre-of-gravity offset, 0.30 m behind the column.** Estimated from the counterweight and pump positions. Affects stability only; does not touch the governing column or weld checks.
- **Front overhang to tipping line, 1200 mm.** Derived from the base drawing dimension chain. Used for stability and wheel reactions. The warning-plate 420 mm dimension should be reconciled against this on the machine.
- **Cylinder lever arm and rod-end pin.** The cylinder anchor geometry (lever arm 0.346 m) reproduces the issued cylinder force but is not dimensioned on the drawings, and the rod-end pin diameter is to be confirmed (see Sections 7.4 and 12).
- **Material yield.** Nominal S275 (275 MPa) is used for all working-stress checks. Mill certificates or a hardness check would confirm the actual yield, which Section 12 shows is likely higher and would recover column margin.
- **Weld throats, 6 mm.** Taken as the drawing minimum. If site welds are smaller the weld checks worsen.
- **Pin verification basis.** See Section 7.4. The reviewer should confirm whether the conservative series-factor basis or the conventional lifting-pin basis applies to the pin verdicts.

15. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion Files

- MATLAB R2026a parametric script (jib_crane_verification.m), single self-contained file driving every number in this report
- Console log (console_log.txt), full text of the run including the factor-of-safety table and the proof-load reconciliation
- Figures 1 to 10 as PNG, and the operating-envelope animation as MP4
- workspace.mat, saved variables for re-plotting without re-running

Appendix B: Result Index by Figure

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Fig 6.2	Wheel reactions and outrigger stress
Fig 7.1	Boom bending moment and stress
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Fig 8.2	Frame stress distribution
Fig 9.1	Operating-envelope sweep
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Fig 11.1	Factor of safety summary